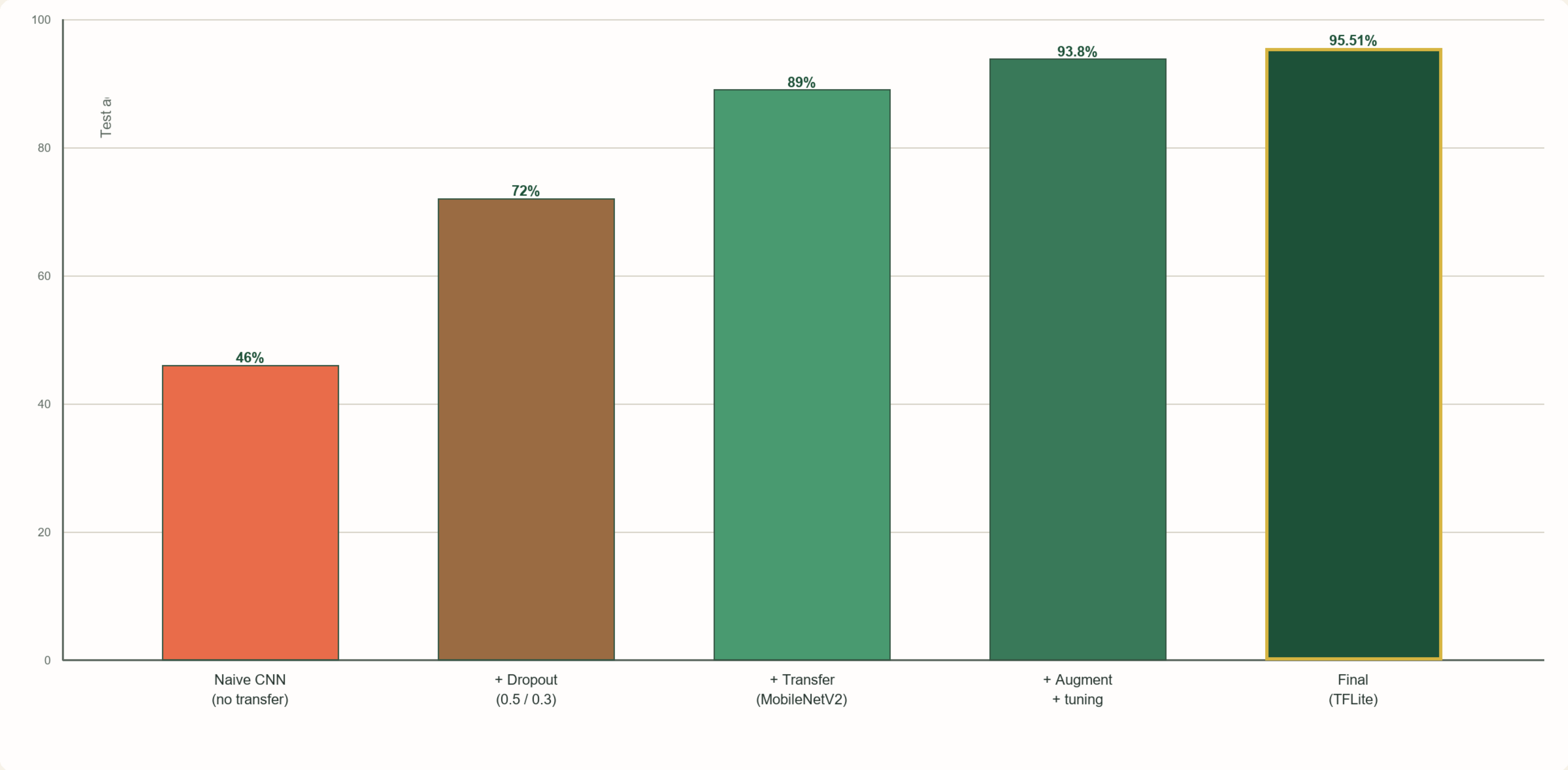
DISCUSSION & IMPACT

ABLATION STUDY

We iteratively added techniques to the plant model. Each step was evaluated on the same PlantVillage test split to show how disciplined model improvements translated into deployable accuracy gains.



+ 49.5 percentage points from v1 to final through systematic iteration.

CHALLENGES OVERCOME

#1 Overfitting on PlantVillage
Problem: Train accuracy greatly exceeded validation accuracy.
Solution: Added dropout (0.5 / 0.3) between dense layers.

#2 No real insect dataset
Problem: Public pest datasets were small and inconsistent.
Solution: Built a synthetic generator with 1000 images across 15 classes.

#3 Motorola TFLite incompatibility
Problem: Deployment device would not load the .tflite model.
Solution: Re-ran export with TFLITE_BUILTINS and fixed Gradle dependencies.

#4 Arthropod visual ambiguity
Problem: Insect classes were visually hard to distinguish.
Solution: Used highlighted active regions plus augmentation; 83.33% was reached.

APP FEATURE INVENTORY

1. **Plant Encyclopedia** — Offline JSON + SQLite, image galleries, search, linked from diagnosis.
2. **Regional Guides** — Crop calendars and pest windows keyed by user-selected region.
3. **Satellite / NDVI** — Map tiles plus vegetation vigor views: (NIR-Red) / (NIR+Red).
4. **Soil Moisture** — SMAP / Sentinel-derived tiles, color-coded with confidence.
5. **OpenWeather** — 24-72 hour forecast, frost risk, spray windows, and precipitation.
6. **Space Weather (Kp)** — NOAA SWPC feed; GPS precision alerts when geomagnetic activity is elevated.
7. **Telemetry + Active Learning** — Opt-in confirmed images feed retraining and remain GDPR/CCPA aware.
8. **Notifications & Reports** — Push alerts for outbreaks plus weekly or monthly PDF field reports.

RESPONSIBLE AI DESIGN

- **Confidence Gating** — Predictions below 50% softmax are suppressed and the user is prompted to retake the image.
- **No Chemical Dosages** — The chatbot refuses pesticide-dosage requests and defers those decisions to agronomists.
- **Privacy by Default** — Photos stay on-device and telemetry remains opt-in with a clear retention policy.
- **Expert Handoff** — Every advisory response includes a disclaimer and local extension links.
- **No Secret Keys** — The GPT API key is held only on the Render backend and is never embedded in the APK.

PROJECTED IMPACT

10x faster — diagnosis versus lab testing

0 marginal cost — per additional farmer reached

~475 M smallholder farms — worldwide (FAO); 84% under 2 ha

CONCLUSIONS

- H1 supported: MobileNetV2 plus transfer learning plus dropout reached 95.51%, competitive with cloud CNNs while still deployable as a 3 MB TFLite model.
- H2 partially supported: the synthetic-data insect classifier reached 83.33%; curated real pest imagery would likely close the remaining gap.
- A unified offline-first architecture combining inference, advisory logic, and agronomic context is achievable on commodity Android hardware.
- Multi-modal integration of vision, NDVI, weather, and LLM guidance turns a class label into a more actionable farmer workflow.

FUTURE WORK

- Replace the synthetic insect set with curated real pest imagery.
- Use quantization-aware training to further reduce model size.
- Partner with USDA extension offices and NGOs for field validation.
- Expand toward crop-yield forecasting using satellite and weather layers.

REFERENCES (MLA)

- Mohanty, Hughes, and Salathe. "Using Deep Learning for Image-Based Plant Disease Detection." Frontiers in Plant Science, vol. 7, 2016.
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- FAO. Impact of Pests and Diseases on Global Food Security. FAO, 2021. USDA Plant Disease Program.